

Listening to cities

Toward pleasant soundscapes in built environments

29 April 2022

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NST Young Scientists Fellowship

Department of Building Energy Research

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The Republic of Korea



아주대학교
AJOU UNIVERSITY

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⇒ Architecture-medical convergence research with architectural acoustics approaches

4 Targets and Plan in Coming Three years to Ajou University

⇒ Research with Ajou University

⇒ Education with Ajou University

Brief Introduction

Education

- B.Sc.: Architectural Engineering, Hanyang University, Korea (2016)
- Ph.D.: Architectural Environmental Engineering, Hanyang University, Korea (2022)
 - Thesis: **Soundscape Psycho-physiological Restoration with Audio-visual Interaction in Urban Soundscape Experience**

Employment

- 2022 - Present: **NST Young Scientists Research Fellowship**
 - Department of Building Energy Research, The Korea Institute of Civil Engineering and Building Technology (KICT)

Awards and Grant

- 2022: **Ph.D. Excellent Paper Award** (Hanyang University)
- 2020: **Young Professional Award** (International Institute of Noise Control Engineering, i-INCE)
- 2019: **EAA Best Paper and Presentation Award & Young Scientist** (European Acoustics Association, EAA)
- 2019: **Top 10 Best Poster Award** (Nederlands Akoestisch Genootschap, NAG)

Publications

- **21** research articles in SCI(E) / SSCI (**19** Main author) in Soundscape, Environmental and Building acoustics
- **34** international conference papers, **34** Domestic conference papers, **5** Patents , **1** Technology transfer

Certificates

- **LEED AP** Building Design + Construction (**BD+C**), **LEED AP** Interior Design + Construction (**ID+C**), **WELL AP**

Research Topic and Focus

1 **ICT VR-AR-CR** based indoor and outdoor environmental quality evaluation

Multi-sensory perception (audio-visual), Soundscape design, Indoor environment quality, Psychoacoustics
3D audio localization, VR-AR-CR sound rendering system, Space acoustic construction, etc.

***7** research articles in SCI(E), **2** patents

2 Eco-friendly **smart green city** and sustainable **zero energy built environment** design

Deep learning, Machine learning, Carbon-neutral building (net zero building), Human-centered techniques
Public energy data platform, GIS visualization, Purpose built vehicle (PBV), Human behavior, etc.

***1** research articles in SCI(E)

3 **Architectural acoustics** design and **building noise** reduction techniques

Multi-purpose hall, Sound diffusion, Open-plan office, Work productivity, Speech privacy, Absorption materials
Floor impact noise, Water-supply and Drainage noise, Road traffic noise, Environmental noise, Mapping, etc.

***10** research articles in SCI(E)

4 **AI-architectural-medical convergence** with human health monitoring sensors

Digital therapeutics, Mental healthcare, Multidisciplinary approaches, Public health, AI model (CNN, RNN),
Psycho-physiological response (heart rate variability [HRV], brain-wave [EEG], Eye-tracking), etc.

***3** research articles in SCI(E), **3** patents, **1** technology transfer (TT): 50,000,000 KRW

My Research Field “Soundscape”

► Noise environment in urban cities



My Research Field “Paradigm Shift”

► Noise control vs. Soundscape



Annoying
Environmental Sound
(Negative)

Sound
Masking

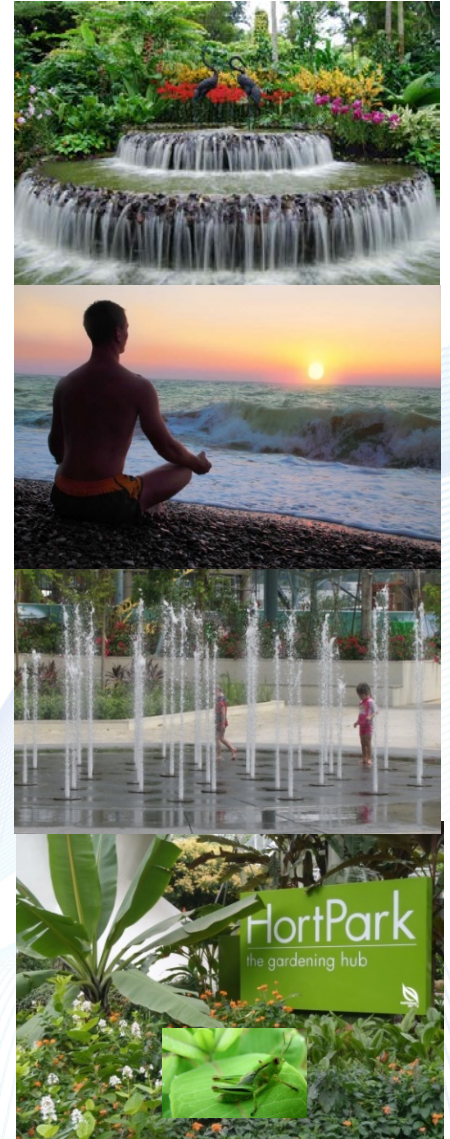
Pleasant
Environmental Sound
(Positive)



Waste
(Reduce)

Resource
(Increase)

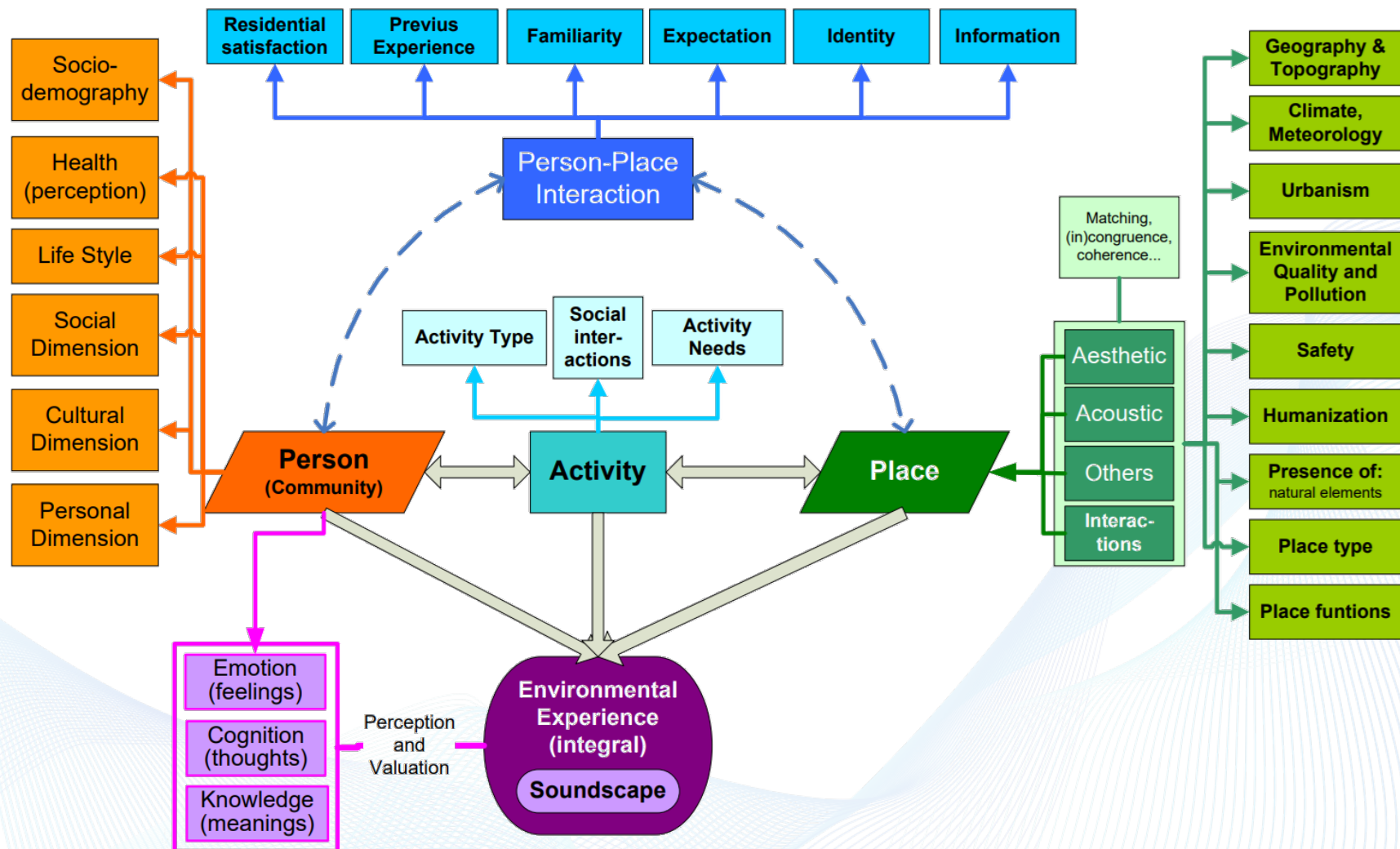
Soundscape Approaches



My Research Field “Context”

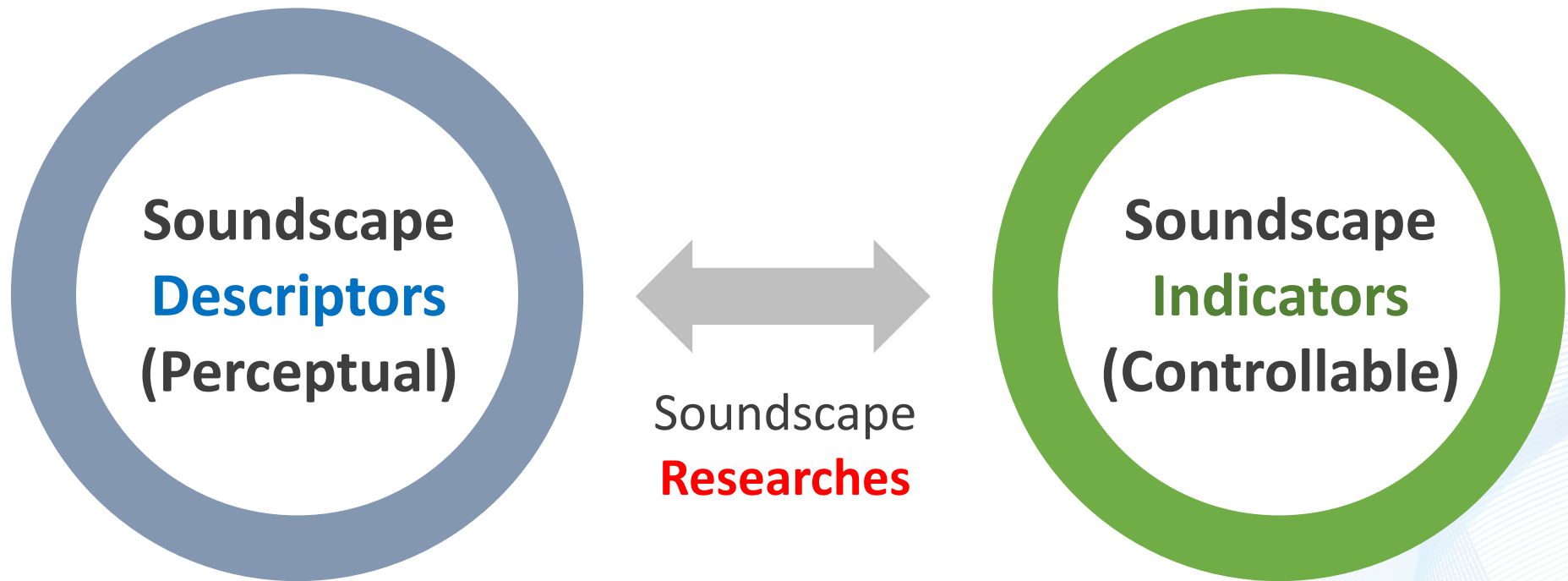
► Soundscape concept (ISO 12913-1, 2014)

- The acoustic environment as **perceived** or **experienced** and/or **understood** by a **person** or **people**, in **context**



My Research Field “Evaluation”

► Soundscape descriptors and indicators



- Noise annoyance
- Pleasantness
- Tranquillity
- Music-likeness
- Perceived affective quality
- Restoration
- Soundscape quality
- Appropriateness

- VR/AR/CR
- ICT/IoT
- Deep learning
- Machine learning
- AI

- Acoustical indices
- Psycho-physiological indices
- Context indices
- Design or remedial indices

My Research Field “Final Goal”

► Eco-friendly sustainable urban environment design

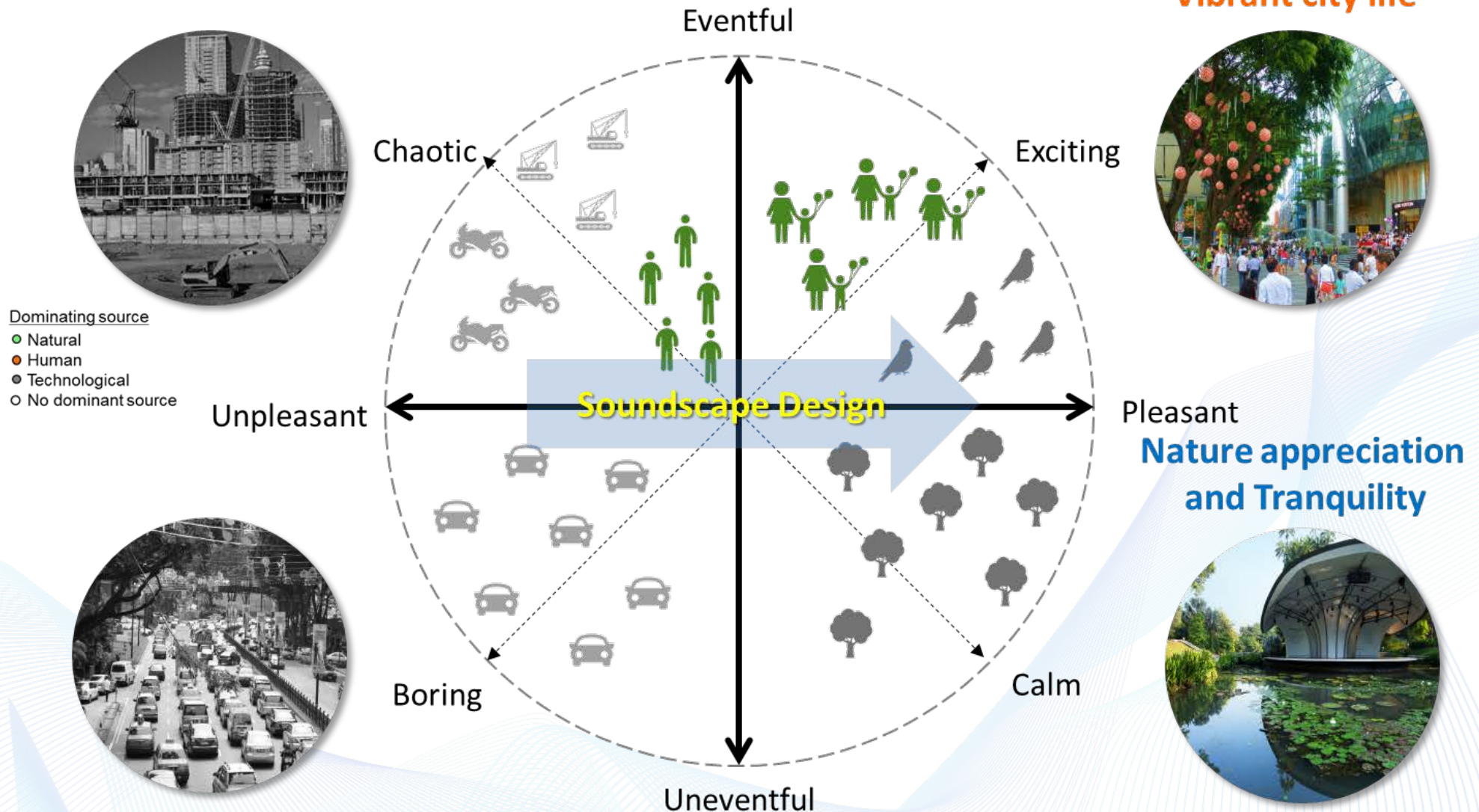


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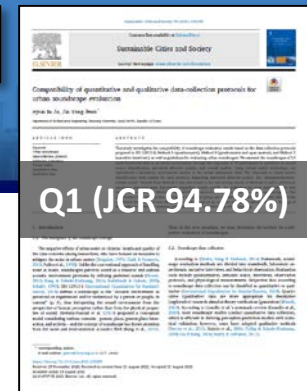
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VR Soundscape Assessment



Joint research institute
University College London
Prof. Jian Kang

Quantitative and qualitative data collection method

- ISO 12913-2: Questionnaire (A and B) and Narrative interview (C)
 - Text-mining analysis and social network analysis (SNA)
- Similar response with Methods A, B, and C



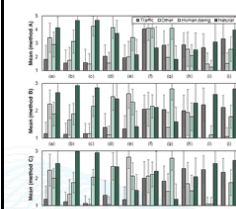
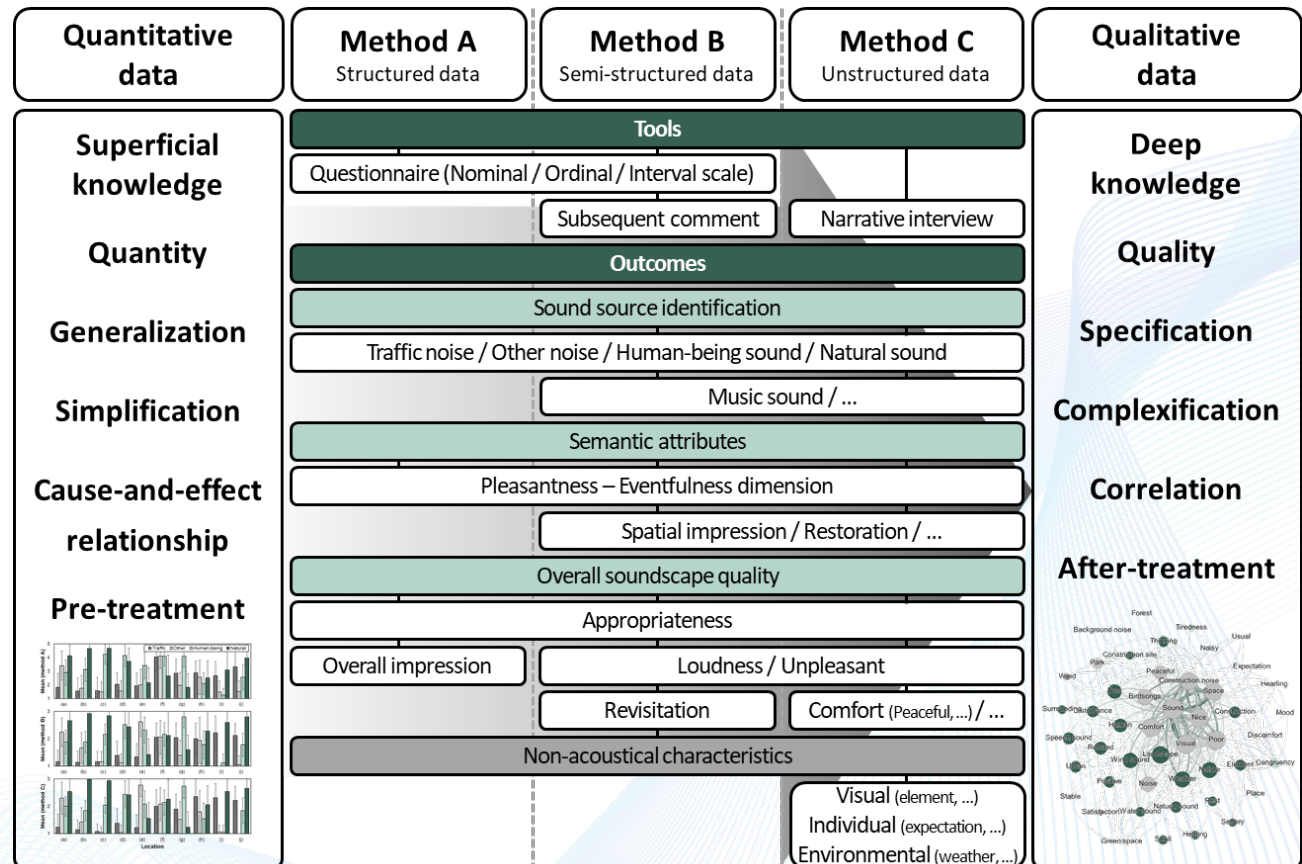
360 Camera



SPS 200



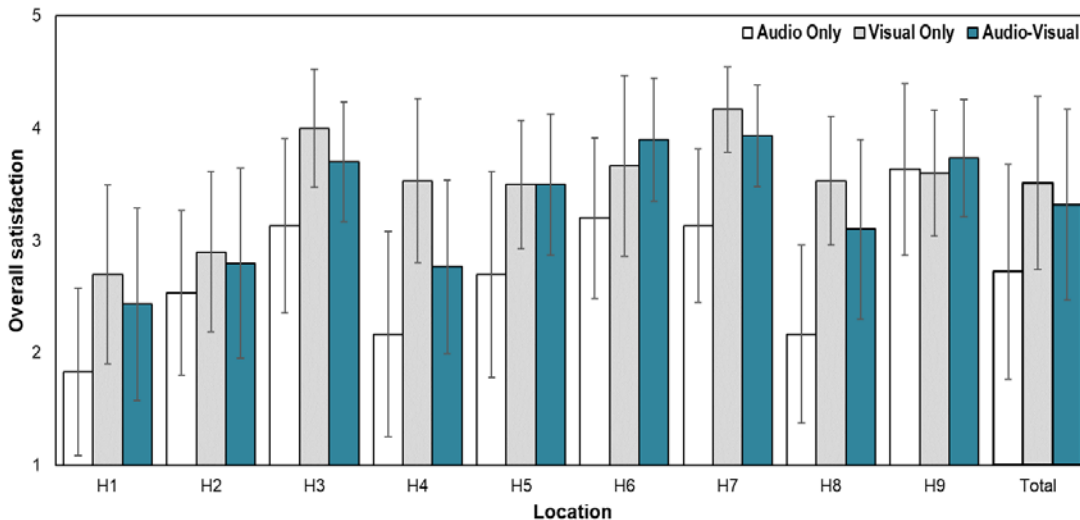
VIVE PRO



Multisensory Soundscape

▶ Audio-visual interaction for satisfaction in smart city

- Satisfaction depending on evaluation environment
 - Visual Only > Audio-visual > Audio Only
 - Contribution: visual information (90%), audio information (10%)



Soundscape dimension

Set-ups	Audio Only		Audio-visual	
	Pleasantness (33.67)	Eventfulness (32.75)	Eventfulness (36.94)	Pleasantness (30.99)
Pleasant	0.78	0.31	0.25	0.84
Unpleasant	-0.82	-0.17	-0.02	-0.84
Calm	0.76	-0.19	-0.38	0.62
Chaotic	-0.85	0.10	0.28	-0.71
Eventful	0.11	0.71	0.86	-0.17
Uneventful	0.07	-0.78	-0.82	0.20
Exciting	0.30	-0.19	0.80	0.32
Monotonous	0.16	0.10	-0.79	0.09

Landscape dimension

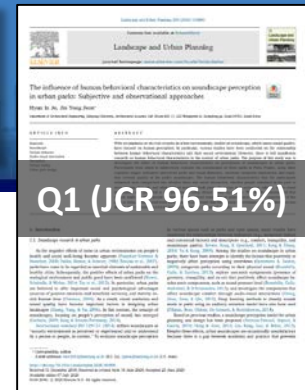
Set-ups	Visual Only			Audio-visual			
	Overall quality (27.66)	Spatial impression (19.02)	Regularity (15.69)	Overall quality (20.33)	Regularity (17.41)	Spatial impression (17.30)	Naturalness (13.40)
Comfortable	0.83	0.14	0.17	0.62	0.38	0.05	0.35
Uncomfortable	-0.73	-0.16	-0.25	-0.61	-0.49	-0.08	-0.20
Harmonious	0.71	0.13	0.31	0.87	0.07	0.09	0.13
Disharmonious	-0.64	-0.01	-0.42	-0.87	-0.10	-0.04	-0.03
Natural	0.80	-0.03	0.01	0.24	0.31	0.02	0.76
Artificial	-0.69	0.09	0.21	-0.10	0.08	0.05	-0.89
Open	0.08	0.74	0.13	-0.01	0.34	0.67	0.13
Closed	-0.02	-0.39	0.52	0.07	0.43	-0.53	-0.09
Wide	0.08	0.87	0.09	0.16	0.07	0.78	-0.04
Narrow	0.01	-0.83	-0.02	-0.07	-0.12	-0.81	0.13
Orderly	0.16	0.20	0.71	0.18	0.78	0.12	0.11
Disorderly	-0.19	-0.19	-0.83	-0.20	-0.81	-0.20	-0.05



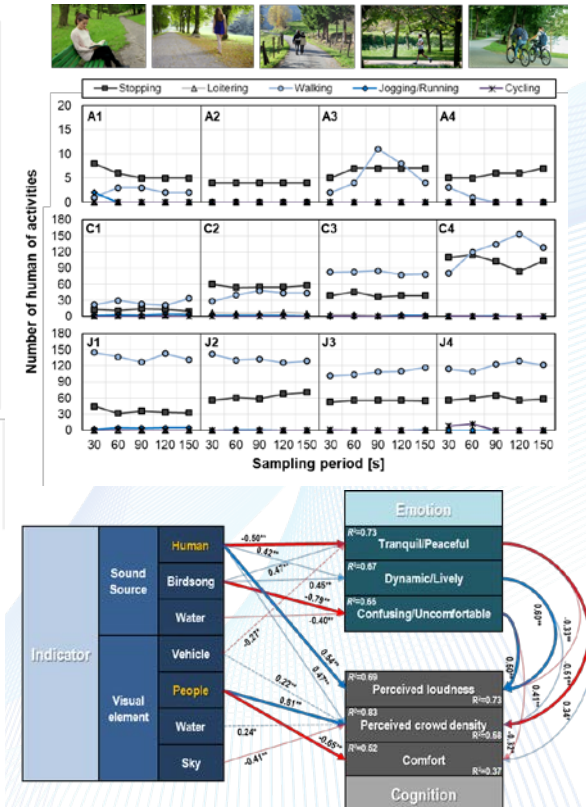
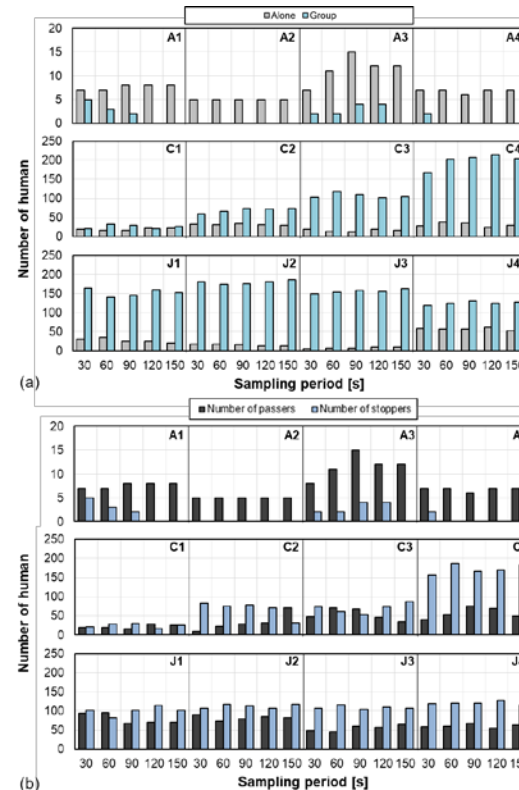
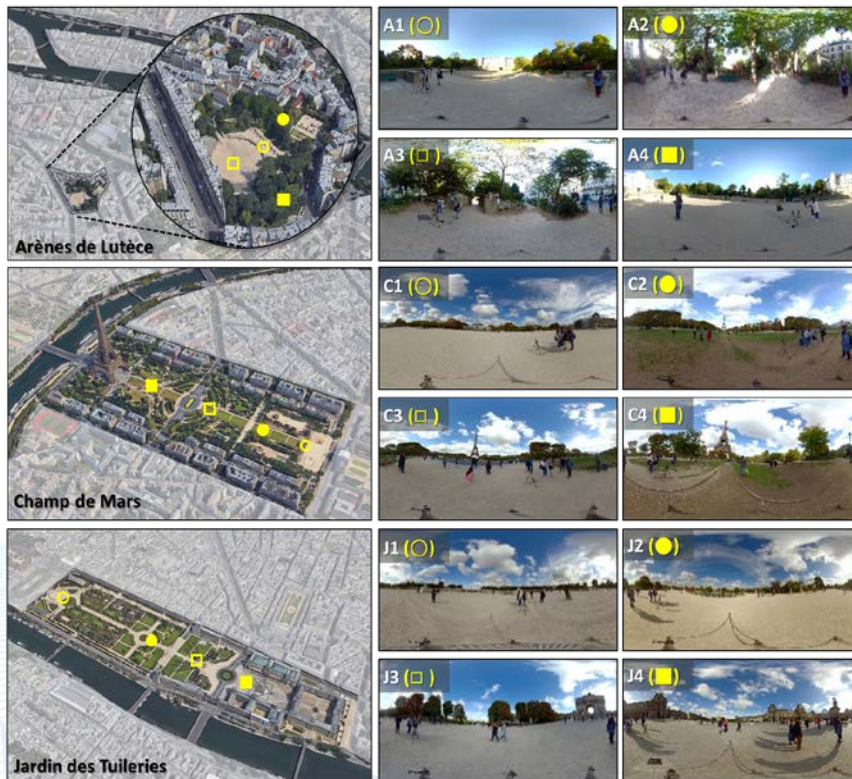
Human Behavior

► Temporal human behavior pattern

- Social interaction, Stay condition, Activity type
- Human presence → Dynamic and lively soundscape
 - Main factor: variations of stopper and passer (activity type not important)



Joint research institute
Sorbonne University
Prof. Dominique-Polack



Smart City Soundscape Design



Joint research institute
Sorbonne University
Prof. Dominique-Polack

Pleasant soundscape design guideline

- Revised soundscape and landscape index
- SEM model considering audio-visual interaction
 - Securing pleasantness, Controlling eventfulness

Reference Index	Definition	Formula	Revised index	Formula
GSI	Green soundscape index ¹⁾	$\frac{\langle PNS \rangle}{\langle PTN \rangle}$	R_GSI	$\frac{\langle PNS \rangle}{\langle PTN \rangle} \times \frac{1}{\langle PL \rangle}$
BGSI	Birdsongs based green soundscape index	$\frac{\langle PBS \rangle}{\langle PTN \rangle}$	R_BGSI	$\frac{\langle PBS \rangle}{\langle PTN \rangle} \times \frac{1}{\langle PL \rangle}$
WGSJ	Water sounds based green soundscape index	$\frac{\langle PWS \rangle}{\langle PTN \rangle}$	R_WGSJ	$\frac{\langle PWS \rangle}{\langle PTN \rangle} \times \frac{1}{\langle PL \rangle}$
RSI	Red Soundscape Index ²⁾	$\frac{\langle PNS \rangle}{\langle PHN \rangle}$	R_RSI	$\frac{\langle PNS \rangle}{\langle PHN \rangle} \times \frac{1}{\langle PL \rangle}$
BRSI	Birdsongs based red landscape index	$\frac{\langle PBS \rangle}{\langle PHN \rangle}$	R_BRSI	$\frac{\langle PBS \rangle}{\langle PHN \rangle} \times \frac{1}{\langle PL \rangle}$
WRSI	Water sounds based red landscape index	$\frac{\langle PWS \rangle}{\langle PHN \rangle}$	R_WRSI	$\frac{\langle PWS \rangle}{\langle PHN \rangle} \times \frac{1}{\langle PL \rangle}$
GLI	Green Landscape Index	$\frac{\langle PNE \rangle}{\langle PVE \rangle}$	R_GLI	$\frac{\langle PNE \rangle}{\langle PVE \rangle} \times \frac{1}{\langle PC \rangle}$
GGLI	Greenery based green landscape index	$\frac{\langle PGE \rangle}{\langle PVE \rangle}$	R_GGLI	$\frac{\langle PGE \rangle}{\langle PVE \rangle} \times \frac{1}{\langle PC \rangle}$
WGLI	Water based green landscape index	$\frac{\langle PWE \rangle}{\langle PVE \rangle}$	R_WGLI	$\frac{\langle PWE \rangle}{\langle PVE \rangle} \times \frac{1}{\langle PC \rangle}$
RLI	Red Landscape Index	$\frac{\langle PNE \rangle}{\langle PPE \rangle}$	R_RLI	$\frac{\langle PNE \rangle}{\langle PPE \rangle} \times \frac{1}{\langle PC \rangle}$
GRLI	Greenery based red landscape index	$\frac{\langle PGE \rangle}{\langle PPE \rangle}$	R_GRLI	$\frac{\langle PGE \rangle}{\langle PPE \rangle} \times \frac{1}{\langle PC \rangle}$
WRLI	Water based red landscape index	$\frac{\langle PWE \rangle}{\langle PPE \rangle}$	R_WRLI	$\frac{\langle PWE \rangle}{\langle PPE \rangle} \times \frac{1}{\langle PC \rangle}$

PNS: Perceived extent of natural sounds, **PTN:** Perceived extent of traffic noise, **PHN:** Perceived extent of human sounds,
PBS: Perceived extent of birdsongs, **PWS:** Perceived extent of water sounds
PNE: Perceived extent of natural elements, **PVE:** Perceived extent of vehicle elements, **PPE:** Perceived extent of people elements,
PGE: Perceived extent of greenery elements, **PWE:** Perceived extent of water elements
PL: Perceived loudness, **PC:** Perceived crowdedness

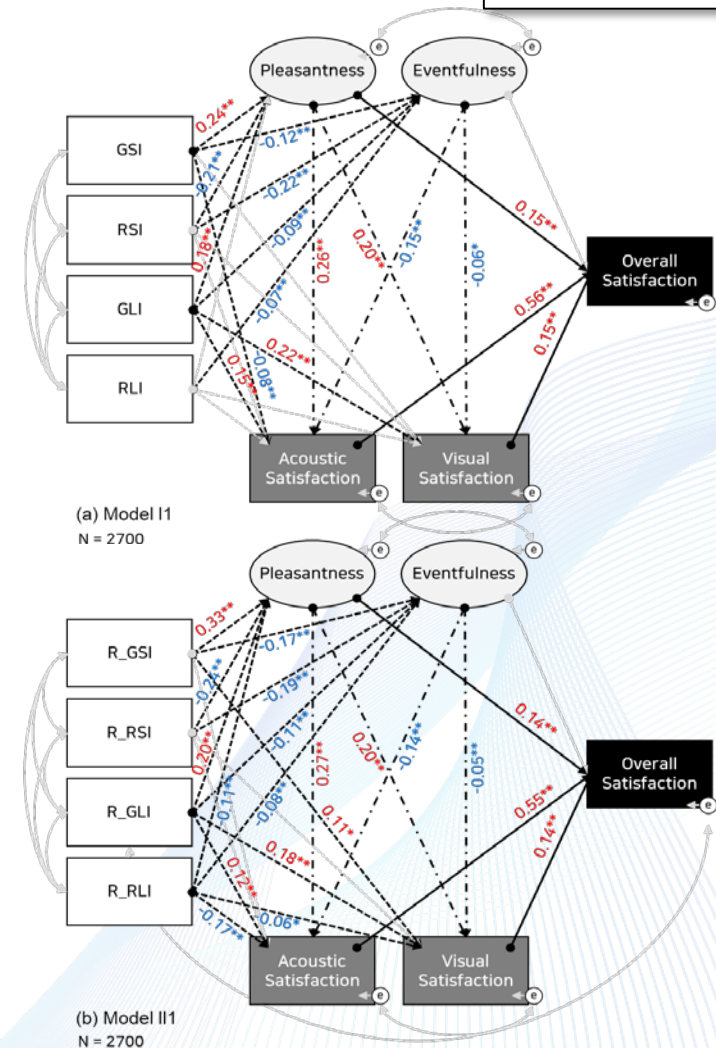


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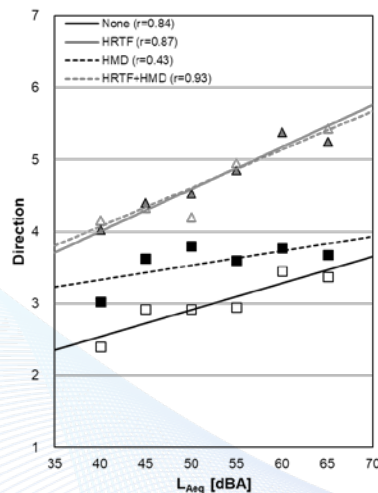
VR Sound Rendering System

HMD + Head related transfer function (HRTF)

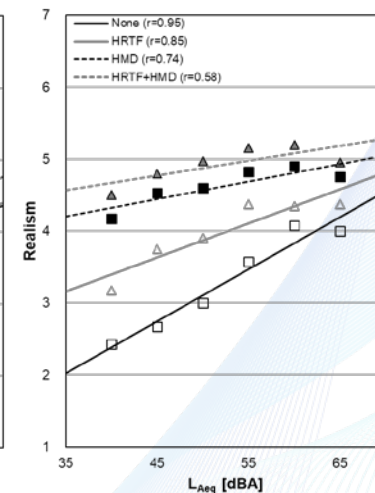
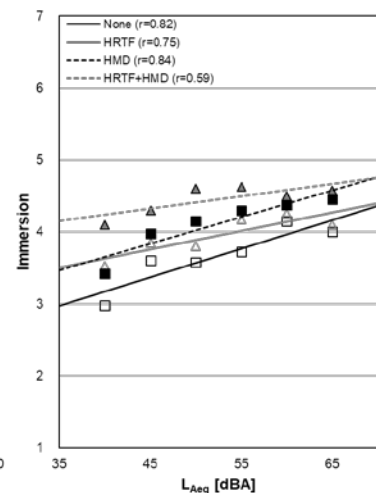
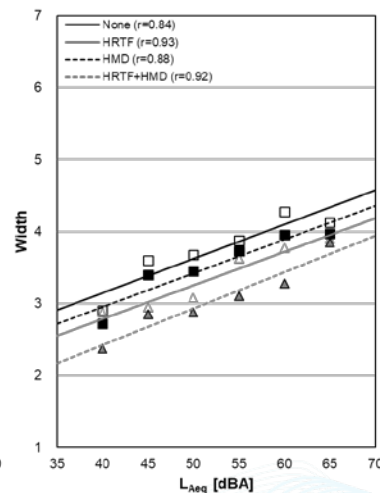
- Effect of HRTF and HMD on subjective evaluation: 77% and 23 %
 - Source-related spatial attributes: HRTF dominant effect
 - Environment-related spatial attributes: HMD dominant effect



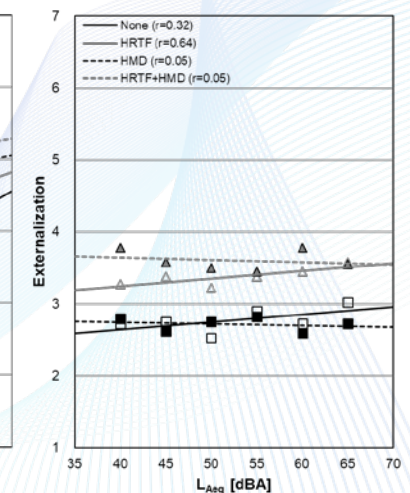
Subjective attributes	Acoustic reproduction methods		
	FOA-static binaural	FOA-tracked binaural	FOA-2D octagonal array
Overall soundscape quality			
• Dominance of sound sources	○	○	○
• Affective quality of soundscape	○	○	○
Source-related spatial attributes			
• Distance	▲	▲	▲
• Directivity	▲	○	○
• Width	○	○	○
• Distinctiveness	▲	○	○



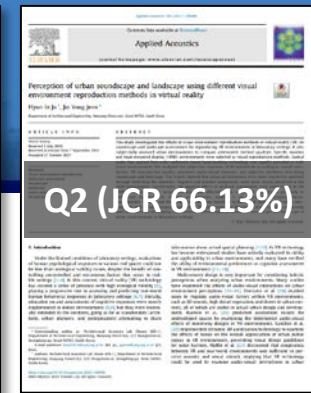
Source-related spatial attributes



Environment-related spatial attributes



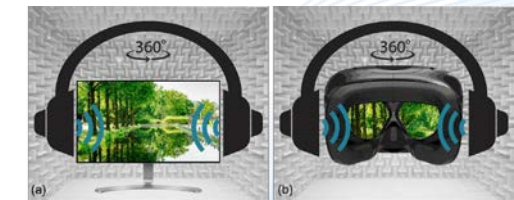
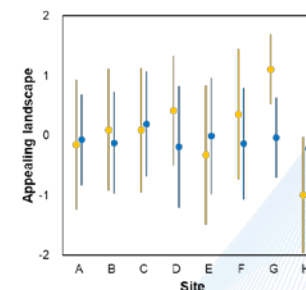
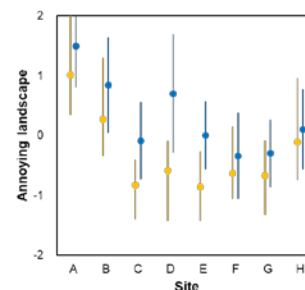
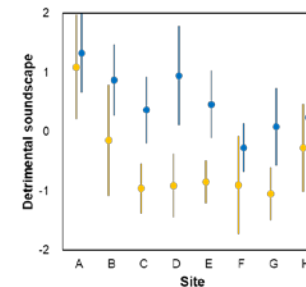
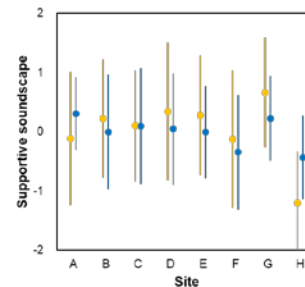
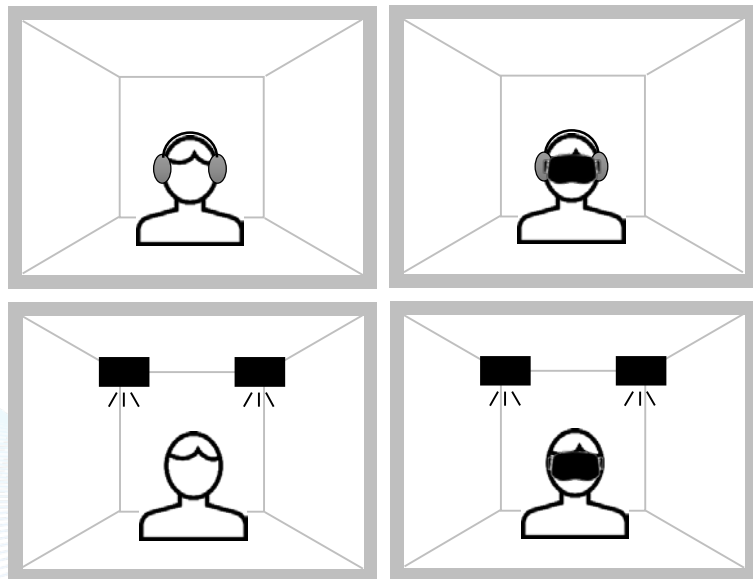
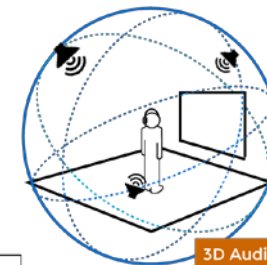
VR Ecological Validity



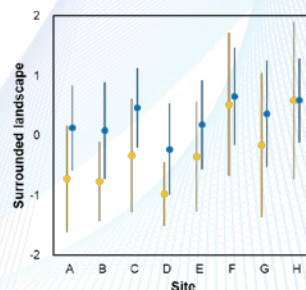
Comparison of audio-visual rendering method

- Headphone vs. Speaker
 - The acceptance limit and noise annoyance: Speaker > Headphone (6 – 8 %)
- HMD vs. Monitor
 - Monitor environments: Prefer to evaluate the overall awareness
 - HMD environments: Prefer to evaluate the spatial presence

Joint research institute
RWTH University
Prof. Micheal Vorlander



● Monitor ● HMD



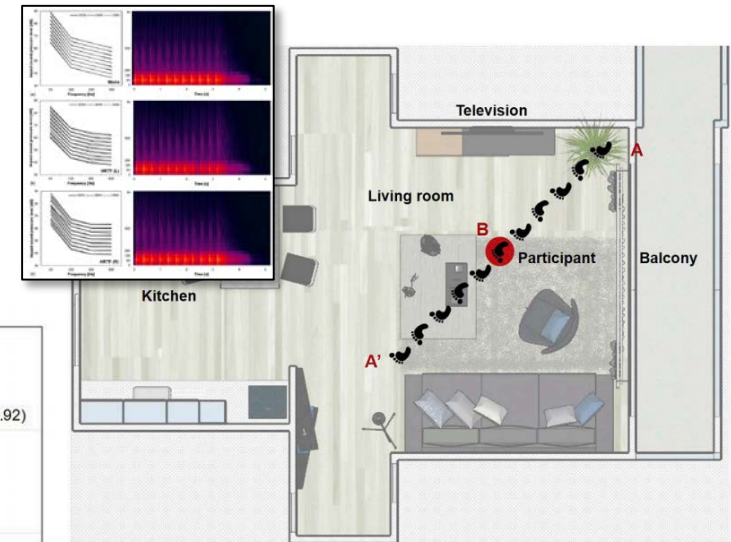
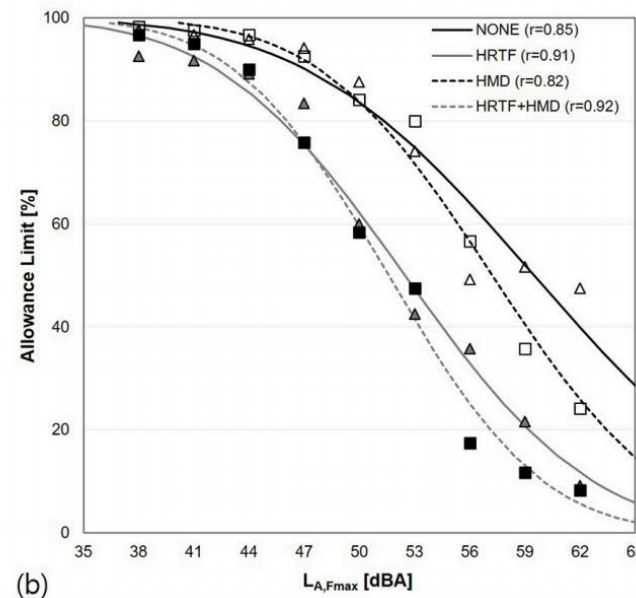
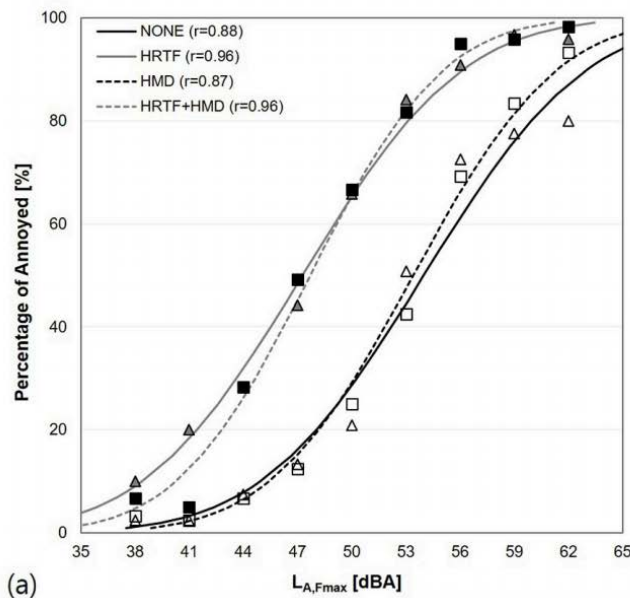
Jeon, J. Y., Jo, H. I.*, Kim, S. M., & Yang, H. S. (2019). Subjective and objective evaluation of water-supply and drainage noises in apartment buildings by using a head-mounted display. *Applied Acoustics*, 148, 289-299. **Cited 11**

Jo, H. I., & Jeon, J. Y. (2022). Perception of urban soundscape and landscape using different visual environment reproduction methods in virtual reality. *Applied Acoustics*, 186, 108498.

VR Application 1

► Noise criteria establishment

- Sound source: Heavy weight floor impact noise
- The %A of heavy-weight floor impact sounds (20% intervals)
- Comparison with existing criteria
 - Lowered the criterion for satisfaction by 6-7 dBA
 - 57 dBA (daytime), 52 dBA (nighttime) -> (III, IV grade)
 - With VR technique -> (IV, V grade)

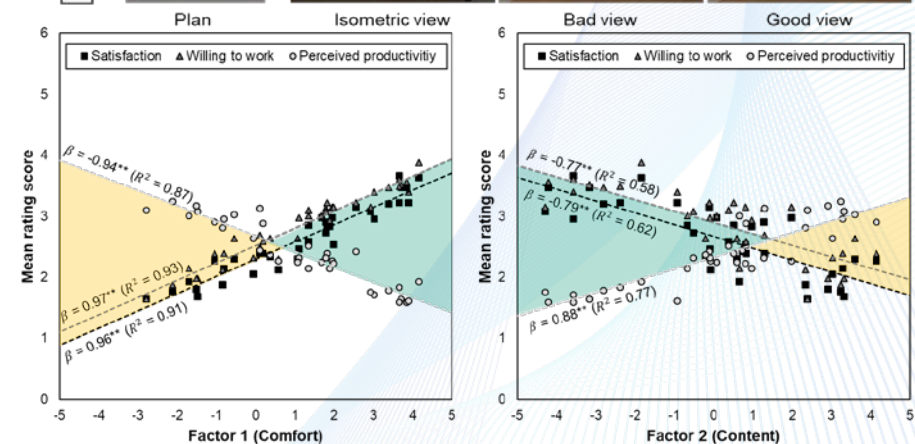
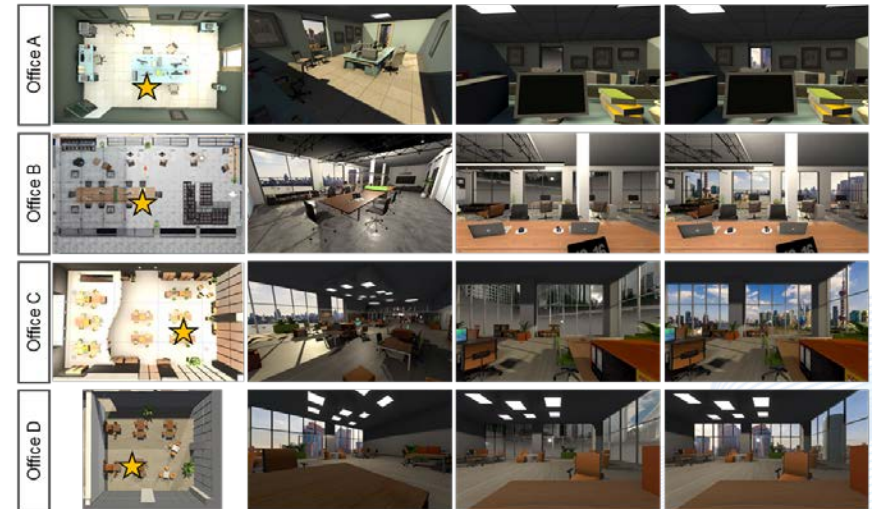
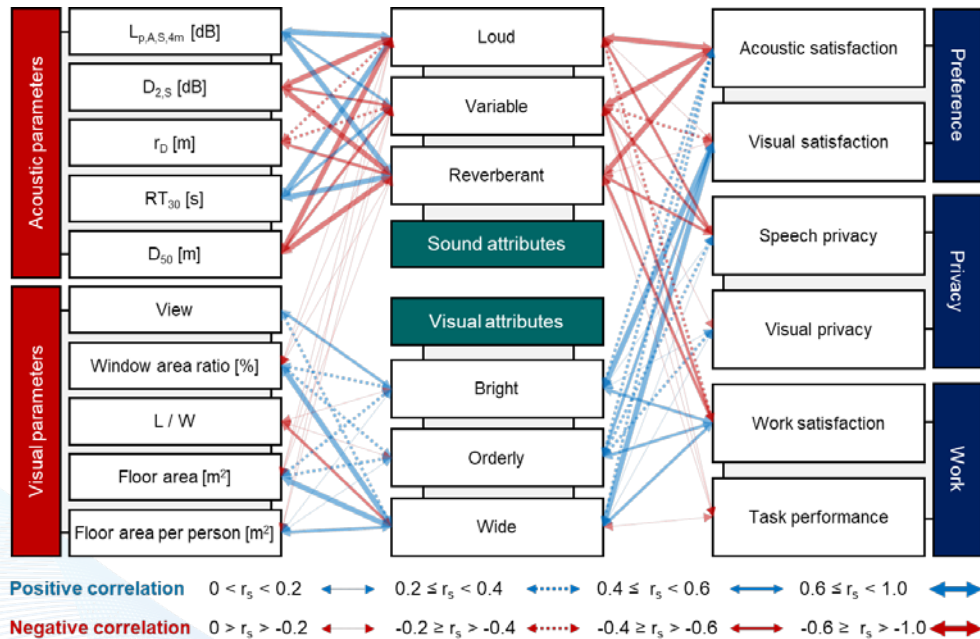


Class	%A	L _{A,Fmax} [dBA]			
		NONE	HRTF	HMD	HRTF+HMD
I	0 – 20%	<48.0	<41.5	<48.0	<42.5
II	20 – 40%	<52.0	<45.5	<52.0	<46.0
III	40 – 60%	<56.0	<49.0	<55.0	<49.0
IV	60 – 80%	<59.5	<52.5	<58.0	<52.0
V	80 – 100%	≥59.5	≥52.5	≥58.0	≥52.0

VR Application 2

Work productivity in open plan office

- Indoor soundscape model (balancing point)
 - Preferred zone: comfort (+), content (-)
 - Productive zone: comfort (-), content (+)
 - Comfort (0.3 – 0.7) and Content (0.6 – 1.2)



Jeon, J. Y., Jo, H. I.*, Santika, B. B., & Lee, H. (2022). Crossed effects of audio-visual environment on indoor soundscape perception for pleasant open-plan office environments. *Building and Environment*, 207, 108512.

Jo, H. I., & Jeon, J. Y. (2022). Influence of indoor soundscape perception based on audiovisual contents on work-related quality with preference and perceived productivity in open-plan offices. *Building and Environment*, 208, 108598.

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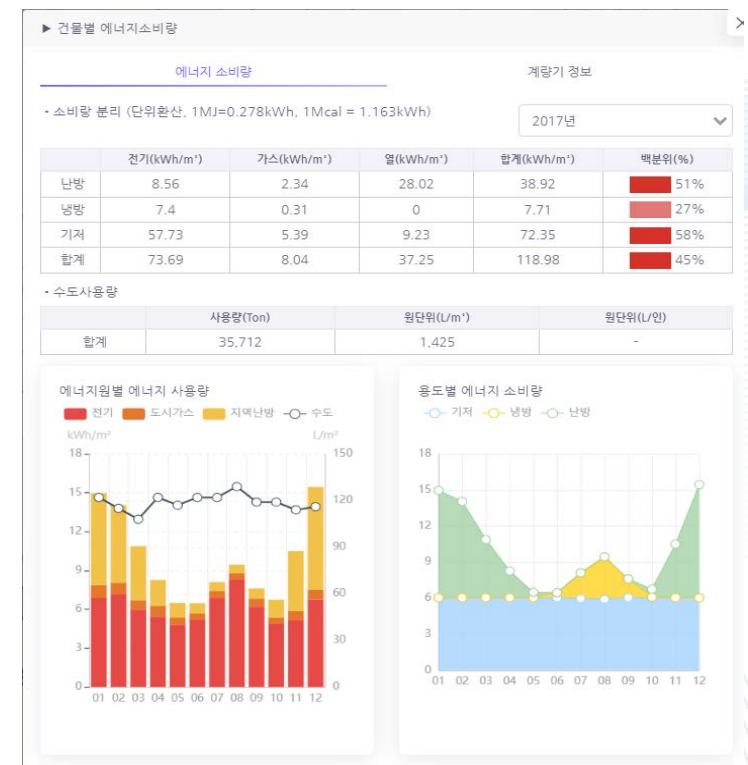
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Data Centric Energy Management

▶ Integrated building energy wide area platform prototype

- Examination filter service (Building usage, energy source, floors, etc.)
- 2D map-based macro-examination (Electricity and gas consumption, etc.)
- 3D map-based micro-examination (Building envelopment, Indoor GML)

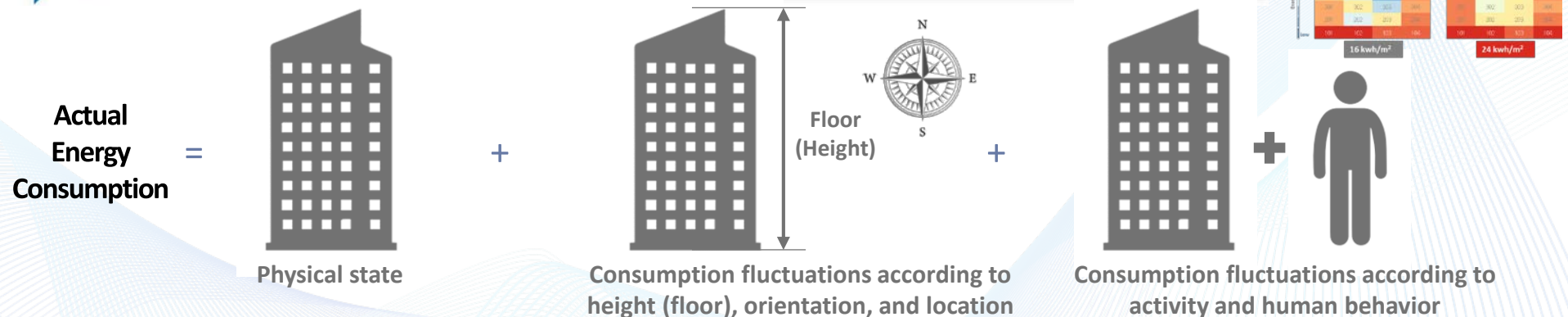
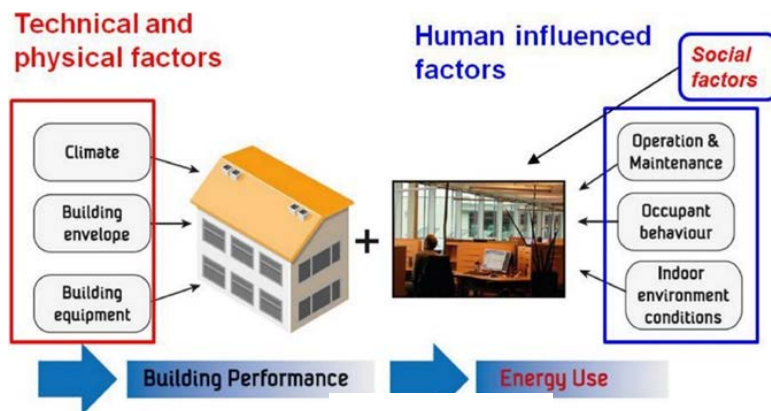


Human Behavior and Energy Usage

► Building energy behavior factors and management indicators

• Building Energy Occupant Behavior Planform

- Physical factors: Building envelope, equipment, climate, shape (GIS information), etc.
- Socio-cultural person: population composition, transportation, geographic information, etc.
- Behavioral Person: Behavior, Operation Management, Survey, IoT Sensing, etc.



4D IEQ Data Visualization Interface

► GIS based Indoor environmental quality auto-visualization

- Questionnaire creation and management interface (K-BOSS)
 - Spatial information-based indoor environment information integrated index
 - Automatic creation of building polygons, space zoning, and heat-map



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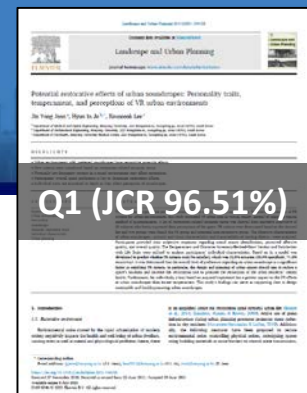
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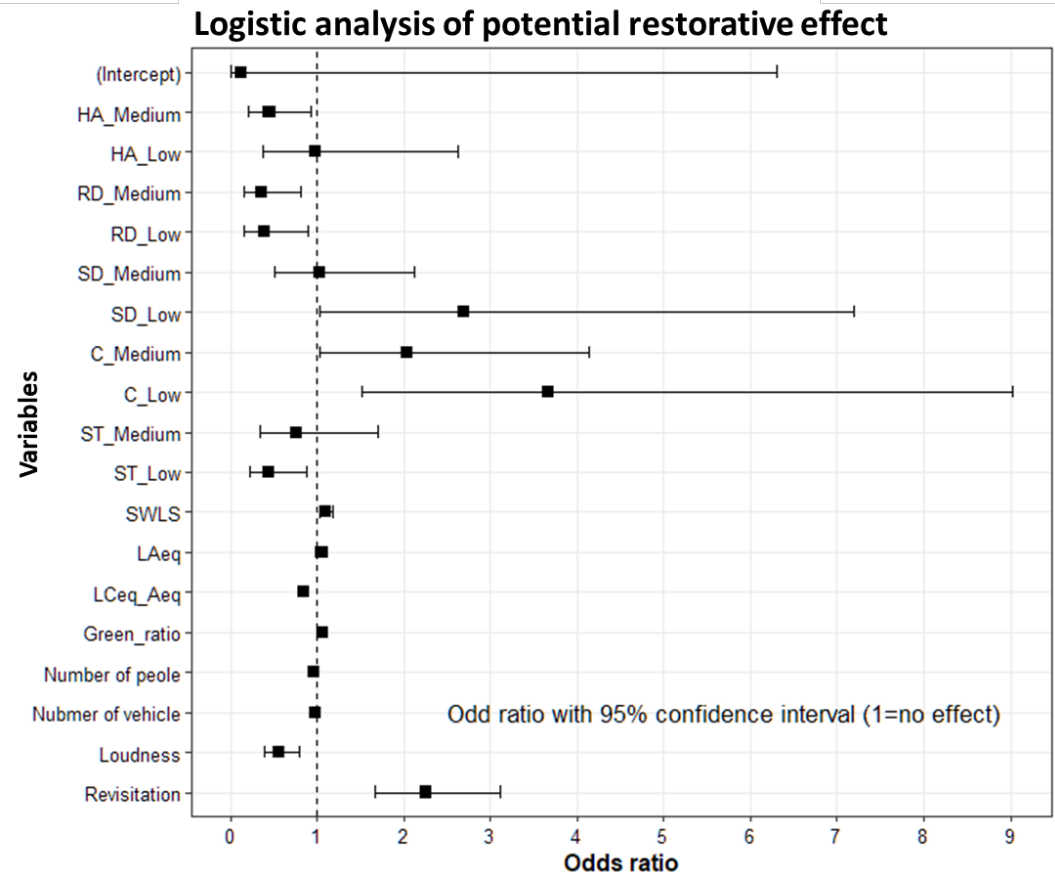
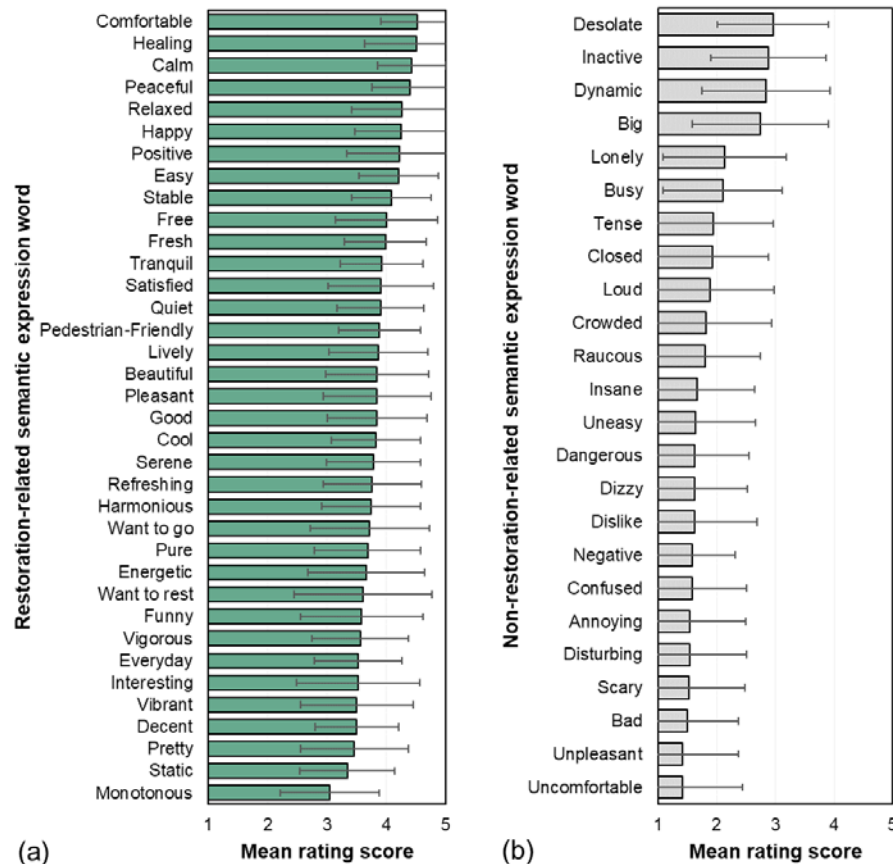
Restorative Environment

Potential restorative effects of urban soundscapes

- Restorative criteria with narrative interview (text-mining analysis)
- Prediction model (Accuracy 82.6 %): Considering individual difference



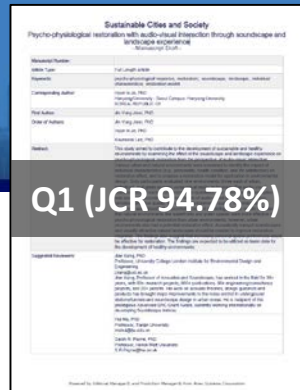
Convergence Research
Architectural Engineering
Psychiatry



Psycho-physiological Restoration

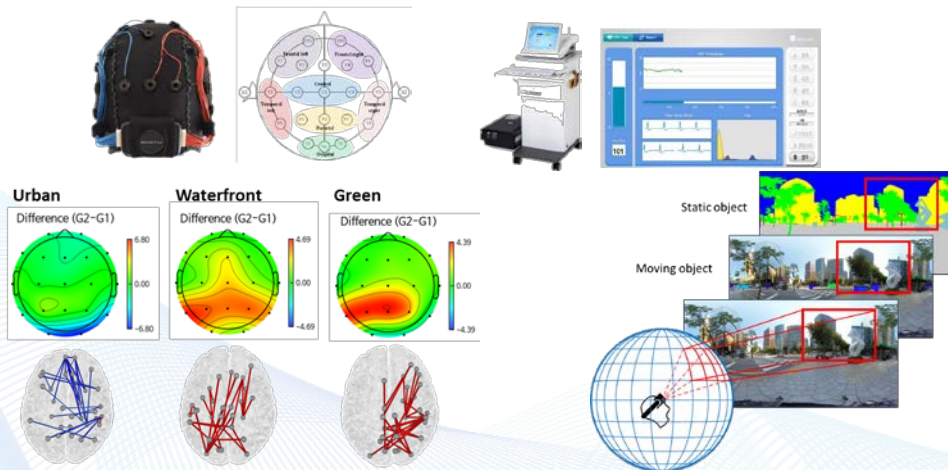
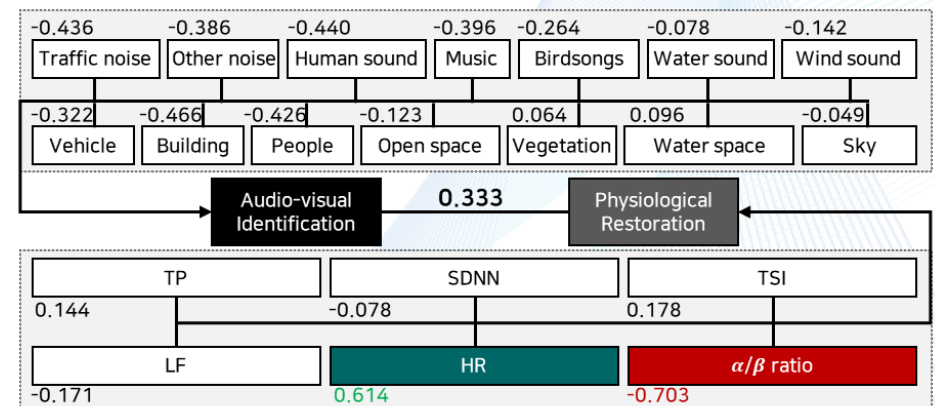
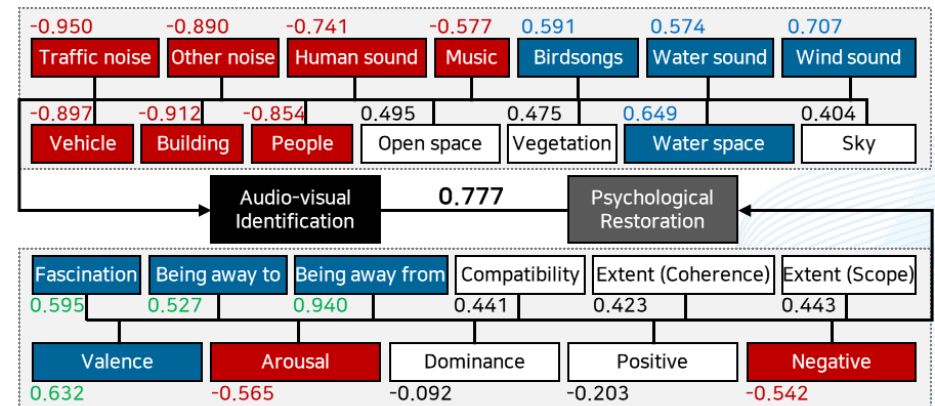
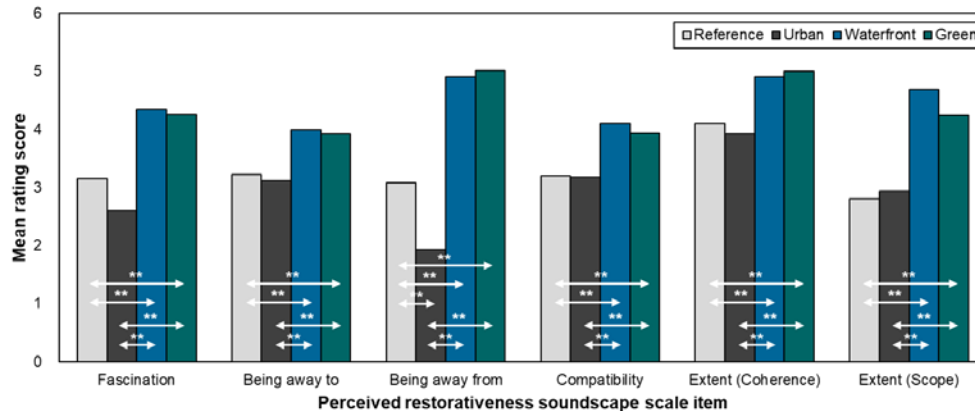
► Multisensory perception and psycho-physiological response

- Physiological response quantification: EEG, HRV, Eye-movement
 - Deep-learning (YOLO and Image segmentation)
- Restoration model (Canonical correlation analysis): **Environment design guideline**



Q1 (JCR 94.78%)

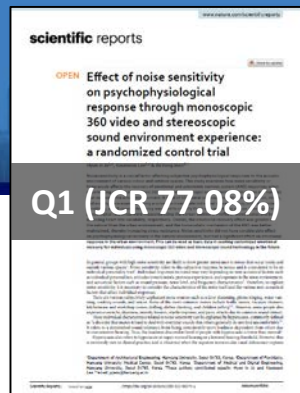
Convergence Research
Architectural Engineering
Psychiatry



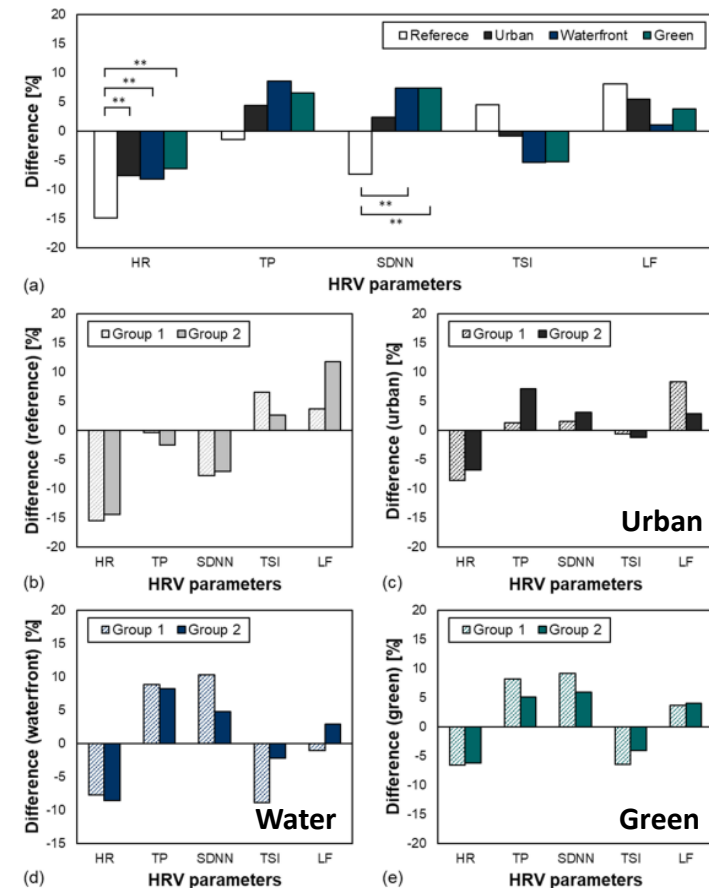
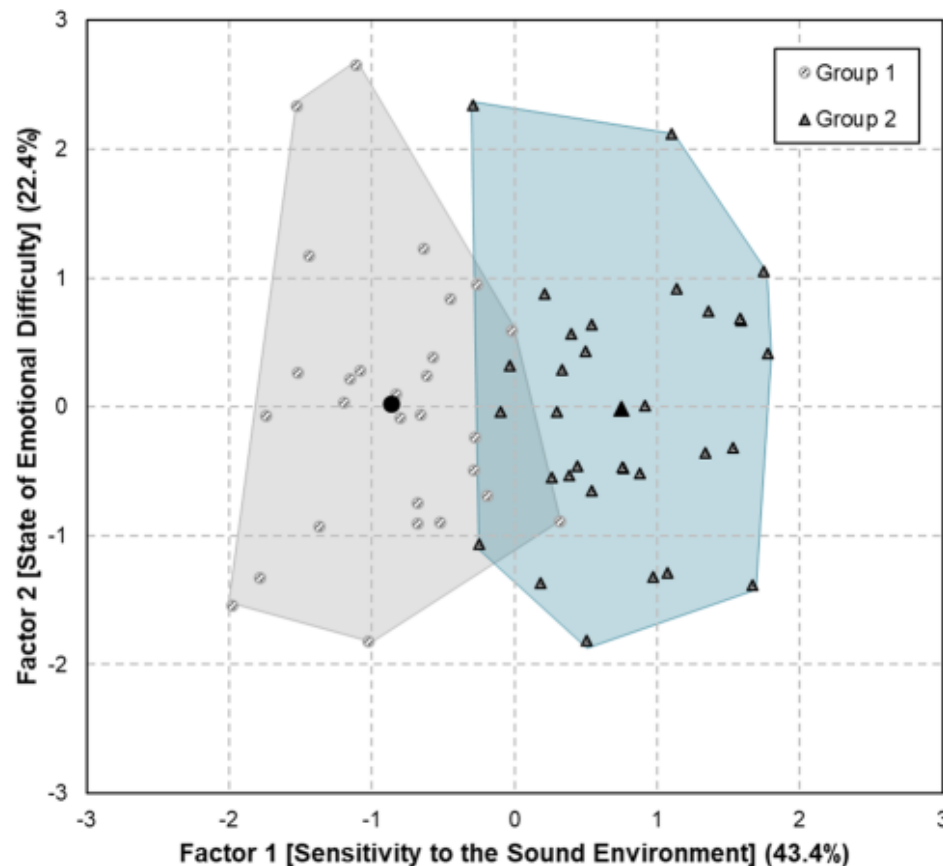
VR Mental Health-care

VR audio-visual environment experience for mental illness

- Alleviating symptom of depression, stress, anxiety
- Main key in physiological restoration: noise sensitivity (in urban)



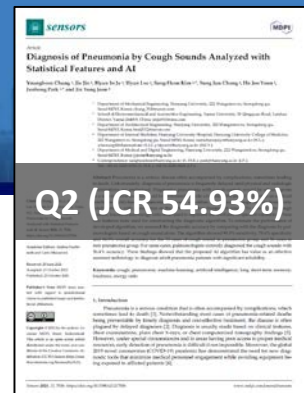
Convergence Research
Architectural Engineering
Psychiatry



Room Acoustic based AI Model

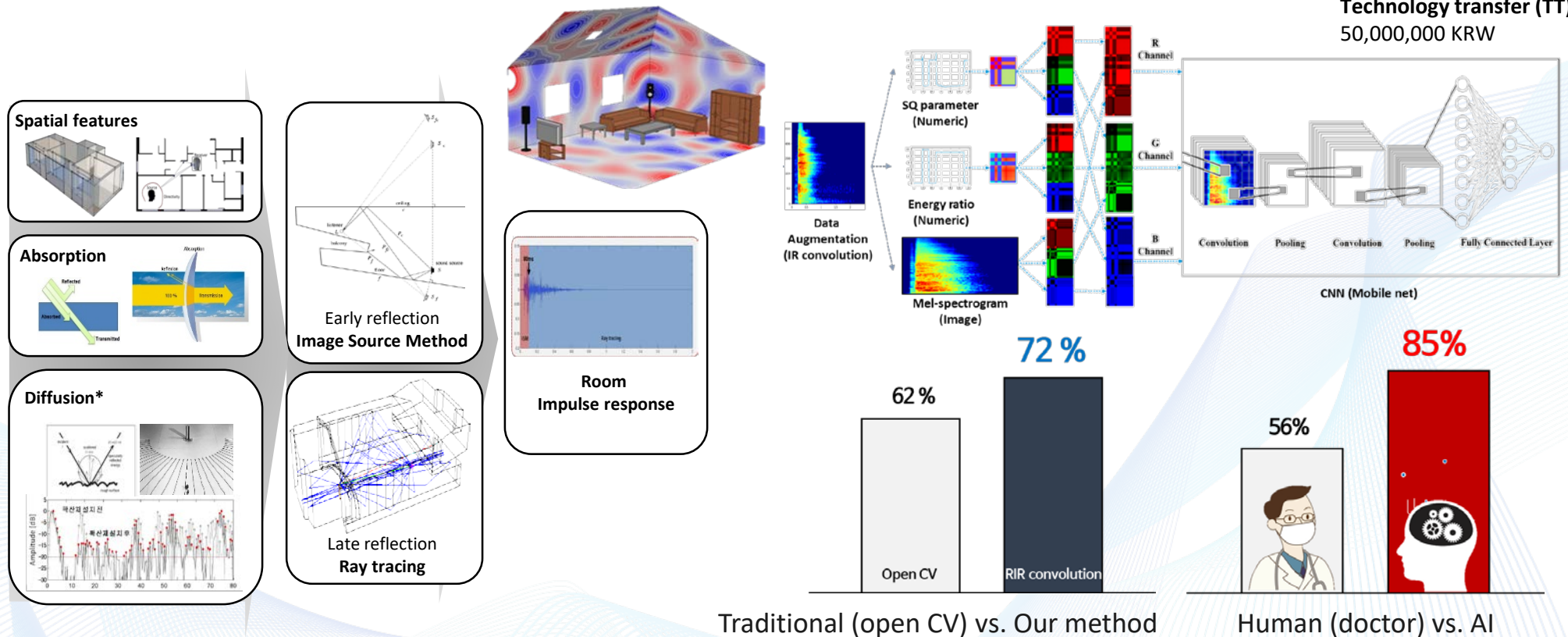
Room impulse response based data augmentation

- Features of respiratory diseases: SQ (Loudness), energy ratio, etc.
- Model: Gramian angular fields (GAF) based CNN
 - Accuracy for diagnosis: 84.9 %



Convergence Research
Architectural Engineering
Mechanical Engineering
Internal Medicine

Technology transfer (TT):
50,000,000 KRW



Summary of Research Topic and Project

1 ICT VR-AR-CR based indoor and outdoor environmental quality evaluation

- Investigation the morphological and behavior factors in urban soundscape (2018-20)
- Development of real-time modeling and rendering methodology of dynamic interactive 3D sound in coexistent space (2016-19)
- Research on sound walking based on spatial sound technology (capturing / noise cancellation / risk analysis) (2017-18)

*7 research articles in SCI(E), 2 patents

2 Eco-friendly **smart green city** and sustainable **zero energy built environment** design

- Data-centric checkup technique of building energy performance (2022-Present)
- Zero energy building support center operation project (2022-Present)

*1 research articles in SCI(E)

3 Architectural acoustics design and **Building noise** reduction techniques

- Office product and space planning guide through sound environment analysis (2019-20)
- Auditory evaluation and questionnaire for water supply and drainage noise (2016-2017)
- Acoustical design of concert hall of Bucheon Culture Arts Center and IFEZ Art Center (2014-22)

*10 research articles in SCI(E)

4 AI-architectural-medical convergence with human health monitoring sensors

- Development of digital therapeutics using social recognition-based VR soundscape for mental illness (2020-22)
- Development of medical solutions with vibro-acoustics (2019-22)

*3 research articles in SCI(E), 3 patents, 1 technology transfer (TT): 50,000,000 KRW

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⇒ My Field of Expertise

3 Most Recently Completed Research Project

- ⇒ Smart green city sound environment design and management for pleasant soundscape
- ⇒ ICT based immersive VR 3D architectural sound environment modeling and rendering
- ⇒ AI Data-centric checkup and optimization technique of building energy performance
- ⇒ Architecture-medical convergence research with architectural acoustics approaches

4 Targets and Plan in Coming Three years to Ajou University

- ⇒ Research with Ajou University
- ⇒ Education with Ajou University

Research Funding Plan

▶ NRF (National Research Foundation) of Korea

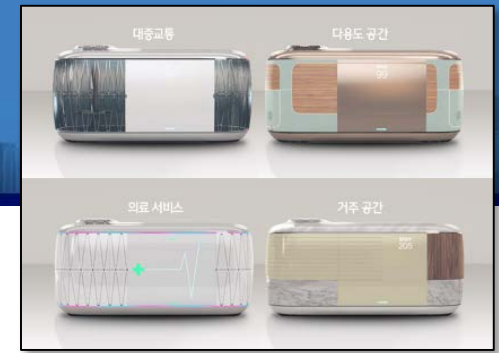
- 우수연구(우수신진연구: 5년 1억), 생애기본연구(생애첫연구: 3년 0.3억원)
- Indoor and outdoor sound environmental design (ICT, VR/AR/MR, AI)
- Title1: Development of integrated soundscape perception model for Human-centered multiscale smart green city sound environment design and management
- Title2: Development of 3D soundscape mapping algorithm based on machine learning
- Low-Carbon sound absorption/diffusion interior materials
- 3D Building noise mapping, monitoring and management

▶ Other funding organization

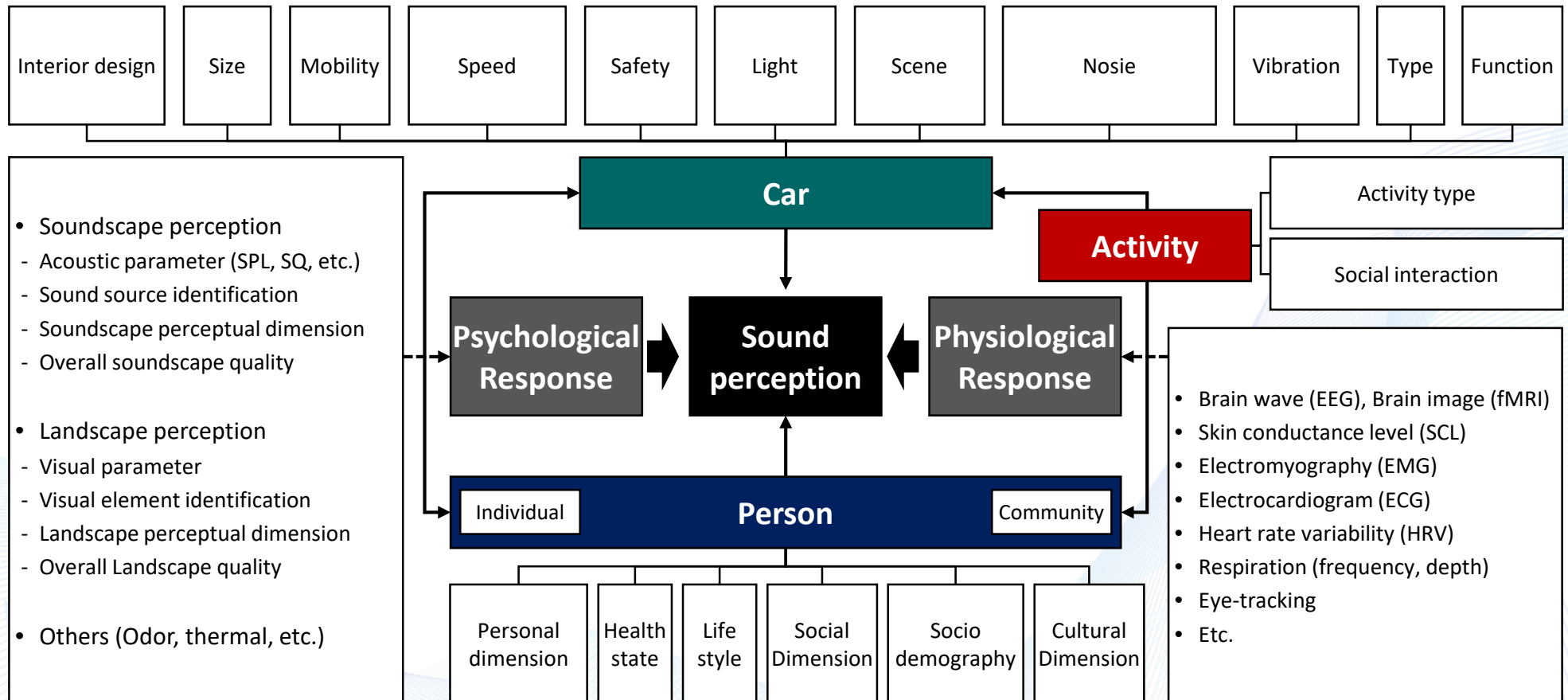
- KICT (The Korea Institute of Civil Engineering and Building Technology)
 - Space acoustics: Sound environment design in space station
- KAIA (Korea Agency for Infrastructure Technology Advancement)
- GBSA (Gyeonggi Business & Science Accelerator)
- KETEP (Korea Institute of Energy Technology Evaluation and Planning)

Industry-University Research

▶ Hyundai joint project “Purpose built vehicle”



- Sound design and research team (2022.09 – 2024.08, 2 years, each 75 million)
 - **Title: Development of Perception Model for Vehicle Sound Environment with User Activity**
 - Vehicle space soundscape evaluation index and comprehensive soundscape perception model

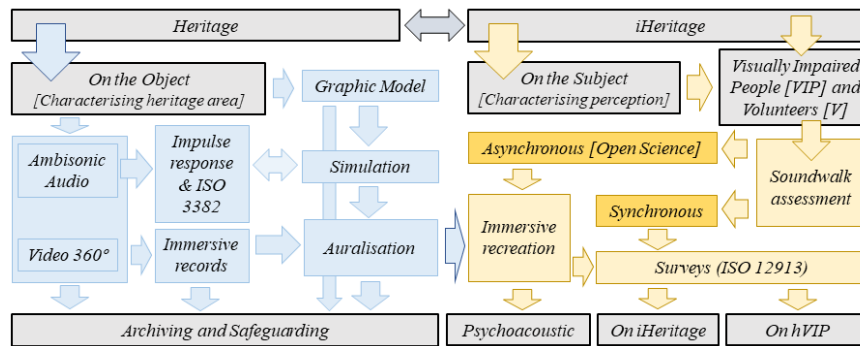


[PBV Soundscape perception model]

International Joint Research

► Horizon Europe (Korea – Europe Research)

- EU-South Korea Collaborative R&D (4 years, each 150 million)
 - **HORIZON- CL2 2022 HERITAGE 01 02 (April 2023; 2ND STAGE, Submitted)**
 - Europe's cultural heritage and arts promoting our values at home and abroad
 - **Title: On inclusive Heritage through Room Acoustics and VIP Soundscape**



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University of Zagreb
Faculty of Architecture



Politecnico
di Torino



- **HORIZON- CL6 2022 COMMUNITIES 02 02 (May 2023; 1ST STAGE, Submitted)**
- Developing nature-based therapy for health and well-being
- **Title: Nature-Based Therapy (walking) in Forest Settings (NateFor)**

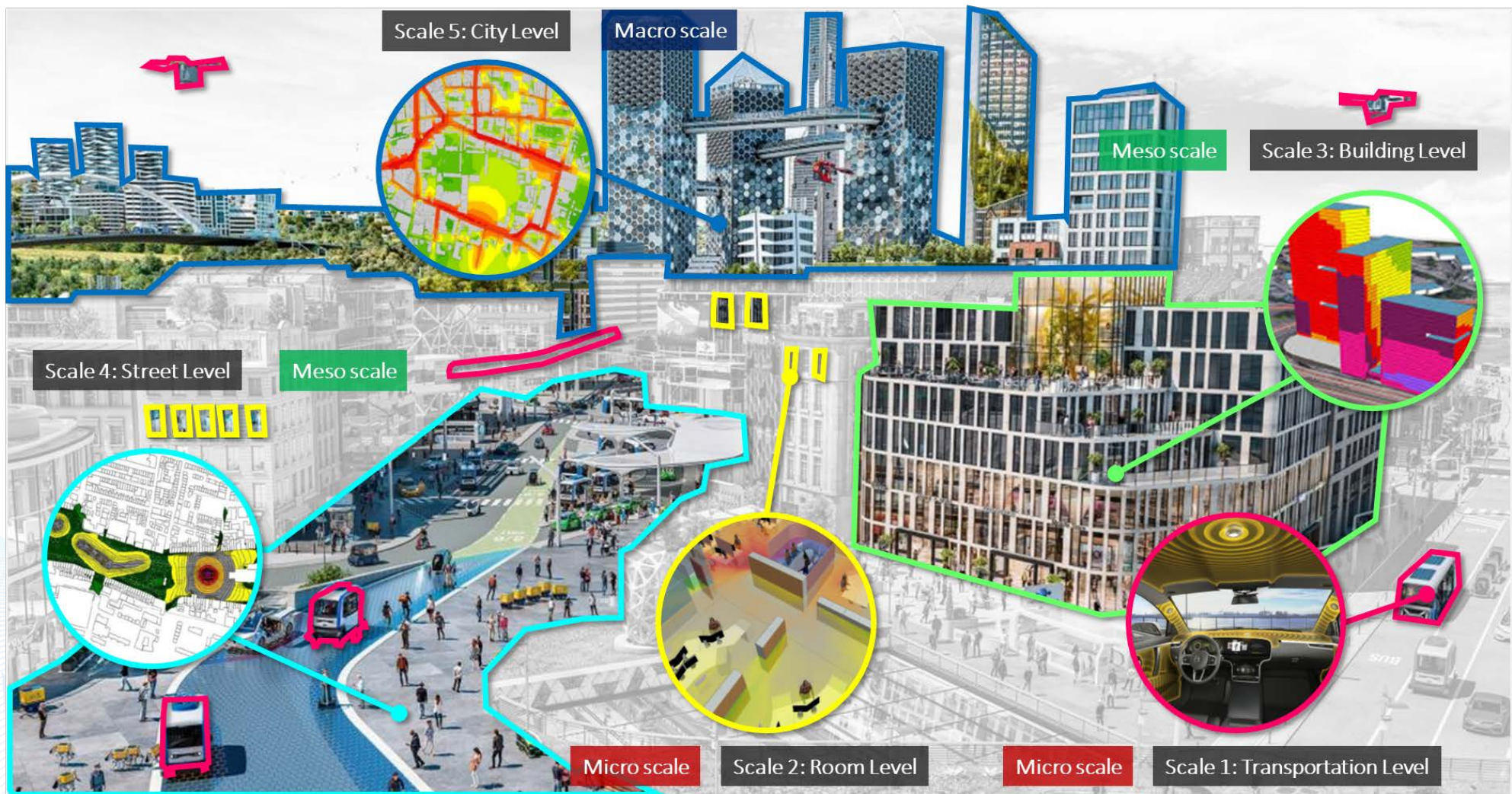
► Cross country joint project

- France-South Korea Collaborative R&D [Sorbonne University]
- UK-South Korea Collaborative R&D [University College London]

Soundscape research (2022 – 2025)

► Human-centered multiscale smart green city sound environment

- Micro-meso-macro scale sustainable and eco-friendly soundscape design



Soundscape research (2022 – 2025)

▶ Micro-level soundscape design (Room and Vehicle)



- Indoor soundscape design index, Psycho-physiological response
- Multi-sensory interaction (sound – visual comfort)
- Method: Deep learning-based sound visualization

*Achievement: 5 SCI(E) papers

▶ Meso-level soundscape design (Building and Street)



- Building façade, Roof garden, Vegetation soundproof wall
- Street canyon aspect ratio, Space function, Human behavior
- Method: VR experience, SEM based perception model

*Achievement: 3 SCI(E) papers

▶ Macro-level soundscape design (Urban)



- GIS-based soundscape mapping (morphological index)
- Natural sound based sound masking system
- Method: Active noise control, ICT based sound monitoring

*Achievement: 2 SCI(E) papers, 1 software (patent)

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4 Targets and Plan in Coming Three years to Ajou University

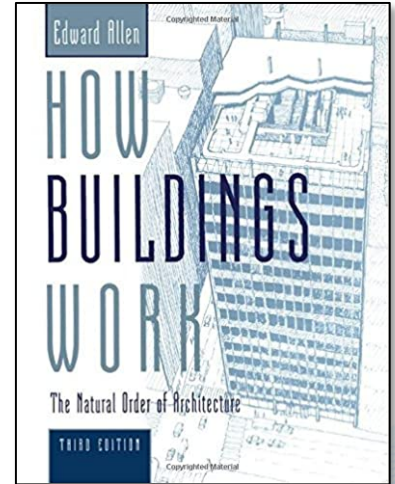
⇒ Research with Ajou University

⇒ Education with Ajou University

Existing Lectures

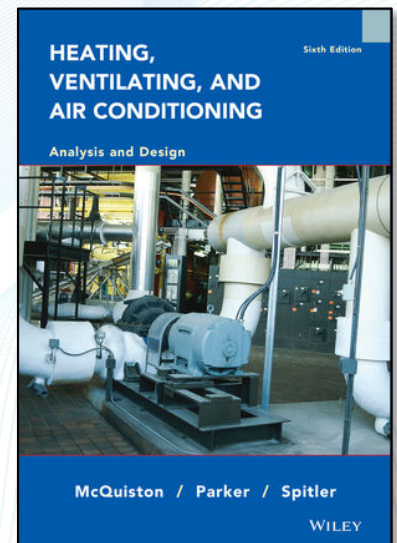
► Architectural Environment

- **Textbook:** How Building Work + Lecture PPT
- **Progression type:** General class + Exploratory class
- **Evaluation:** Attendance (10) + Project (20) + Midterm (35) + Final exam (35)
- Thermal (thermal phenomenon and comfort, heating and insulation)
- Air (air flow and ventilation), Light (natural and artificial lighting)
- Sound (sound field composition and acoustics/noise evaluation), Water
- **Team Project:** Design goal of building (case study, suggesting problem and solution)



► Building Equipment (HVAC)

- **Textbook:** Heating, Ventilating, and Air conditioning + Lecture note
- **Progression type:** General class
- **Evaluation:** Attendance (10) + Report (20) + Midterm (35) + Final exam (35)
- Fundamentals of building mechanical systems
- HVAC system design procedures, codes and standards
- Building environmental control technologies
- **Report1:** Calculation of cooling and ventilation load
- **Report2:** HVAC Control systems design



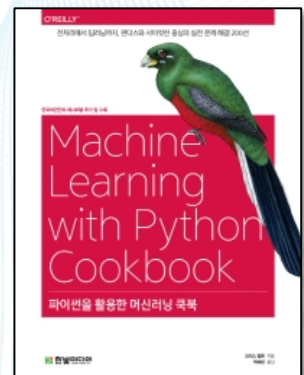
New Lectures for Architect 4.0

▶ Adventure design (Convergence architectural engineering)

- **Textbook:** Lecture note + guest lecture (2 times)
- **Progression type:** Design-oriented class + Industry-Coupled Problem-Based Learning
- **Evaluation:** Attendance (10) + Report (30) + Team project (60)
- Overview of Industry 4.0 technologies (VR/AR, ICT, 3D Printer, Computational design etc.)
- Architect 4.0 industrial revolution technology, industry, and research trends
- Understanding the green and carbon balance based on the IoT and artificial intelligence
- **Team project:** Industry 4.0 techniques idea competition

▶ Building Big data Analysis

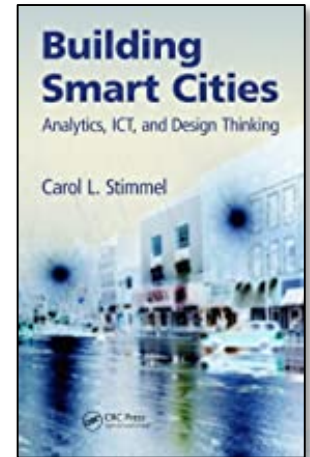
- **Textbook:** Big data architect's handbook + Lecture note
- **Progression type:** Industry-Coupled Problem-Based Learning
- **Evaluation:** Attendance (10) + Midterm (coding test) (40) + Team project (50)
- Basic python grammar (theory of machine learning and deep learning)
- Learn to build and run a big data application with sample code
- Explore examples to implement activities that a big data architect performs
- Use Machine Learning and AI for structured and unstructured data
- **Team project:** Bigdata analysis with open data related building / construction



New Lectures for Built Environment

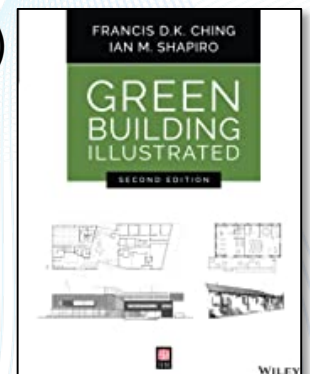
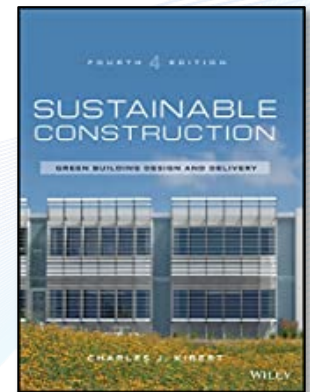
► Introduction to Smart Building

- **Textbook:** Building Smart Cities (Analytics, ICT, and Design Thinking)
- **Progression type:** General class + Exploratory class
- **Evaluation:** Attendance (10) + Report (40) + Midterm (25) + Final exam (25)
- Presentation of the technologies required for coordinated, efficient cities
- Ideation, design, prototyping, and building or retrofitting smart cities
- Proposal of a model for a viable smart city project
- **Report:** Case study of smart cities and smart building



► Sustainable building environment

- **Textbook:** Sustainable construction, Green Building Illustrated
- **Progression type:** General class + Exploratory class
- **Evaluation:** Attendance (10) + Report (30) + Midterm (30) + Final exam (30)
- Get up to date on the green building codes and standards (LEED, WELL, BREEAM)
- Learn about the newest technology in green building materials
- All-new case studies with zero energy performance building
- Expanded coverage of zero energy building design
- Apply sustainability concepts to all aspects of construction and design



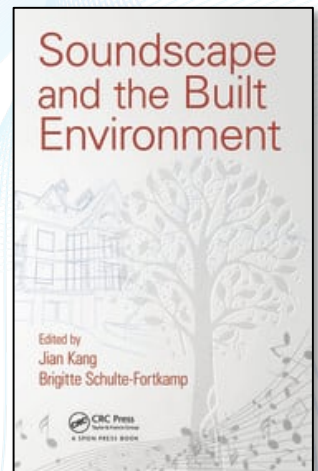
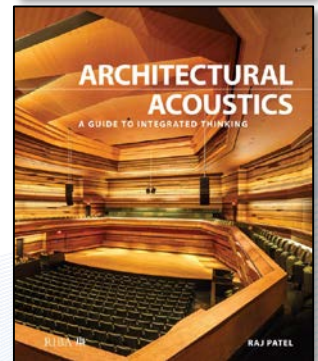
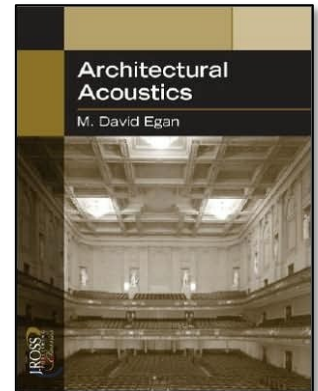
New Lectures for Acoustics

► Architectural Acoustics

- **Textbook:** Architectural Acoustics + Lecture PPT
- **Progression type:** General class
- **Evaluation:** Attendance (10) + Report (10) + Midterm (40) + Final exam (40)
- Basic theory, Sound absorption, Room acoustics, Psychoacoustics
- Noise control, Mechanical system noise and vibration
- Speech privacy and intelligibility, Electronic sound system
- **Report:** Noise reduction method for building noise

► Soundscape

- **Textbook:** Soundscape and the Built Environment + Lecture PPT
- **Progression type:** General class
- **Evaluation:** Attendance (10) + Report (20) + Midterm (35) + Final exam (35)
- Fundamentals of acoustics and soundscape
- Soundscape measurement, evaluation, visualization
- Soundscape design introducing pleasant sounds, landscape ecology
- **Report:** Case study of soundscape design



Thank you for your attention!

Hyun In Jo

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